Database 1

Final Database Project

Third Cut

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Introduction

This document presents the development of an information system for payroll management in Colombia, created for the **Database I** course. The system focuses on optimizing the handling of employee-related information and automating payment processes.

Through this solution, the goal is to ensure accurate salary settlements, including social benefits and social security contributions, in compliance with Colombian regulations. The application relies on a relational database to guarantee data integrity and enable efficient management.

System Modules

1. **Employee Management:**
   - Registration, update, and deletion of employee records (CRUD).
   - Management of salary, contract type, and job position: When an employee is registered in the system, attributes such as salary, contract type, and job position are required.
2. **Payroll Settlement:**
   - **Automated salary calculation:** The system automatically performs calculations based on the data entered at the time of registration.
   - **Application of deductions and bonuses:** The system must allow the input of bonuses and deductions into the database.
   - **Pay stub generation:** The system must generate a payment receipt that can be viewed.
3. **Social Benefits Settlement:**
   - Calculation of severance, severance interest, and service bonus:
     - Severance:
       - Formula: Monthly salary × (Days worked / 360)
     - Interest on severance:
       - Formula: Severance × 0.12 × (Days worked / 360)
     - **Service bonus:**
       - Formula: (Monthly salary × Days worked) / 360
   - **Vacation calculation and settlement:**
     - If the employee has worked for more than one year, they may be granted 2 vacation days.
4. **Social Security and Withholdings:**

- Calculation and registration of contributions to health, pension, and occupational risk insurance (ARL):
  - **Health:** 4% of the employee's monthly salary
  - **Pension:** 4% of the salary
  - **ARL (Occupational Risk Administrator):** 0.5% of the salary
- **Application of tax withholdings:**

  A percentage of the salary is withheld as income tax only if the employee earns more than **$4,500,000 COP per month**.

Salario Mensual	Porcentaje de Retención
Menos de 4.500.000	0% no aplica
4.500.000 - 6.000.000	10%
6.000.000 - 10.000.000	15%
más de 10.000.000	20%

  For a salary of $5,000,000, the withholding would be:

  $5,000,000 × 10% = **$500,000**

5. REPORTS:

- Generation of detailed reports on payments and social security contributions.
- The information could be displayed in a clear and understandable table.